

ASYMPTOTIC EFFICIENCY IN ESTIMATION WITH CONDITIONAL MOMENT RESTRICTIONS

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Received March 1984, final version received May 1986

In this paper, bounds on asymptotic efficiency are derived for a class of non-parametric models. The data are independent and identically distributed according to some unknown distribution F . There is a given function of the data and a parameter. The restrictions are that a conditional expectation of this function is zero at some point in the parameter space; this point is to be estimated. If F is assumed to be a multinomial distribution with known (finite) support, then the problem becomes parametric and the bound can be obtained from the information matrix. This bound turns out to depend only upon certain conditional moments, and not upon the support of the distribution. Since a general F can be approximated by a multinomial distribution, the multinomial bound applies to the general case.

1. Introduction

We shall develop bounds on asymptotic efficiency for a certain class of non-parametric models. For a simple example, assume that $\{(x_{i1}, x_{i2}, y_i)\}_{i=1}^{\infty}$ is independent and identically distributed according to some unknown distribution F_0 , and that $E(x_{i1} u_i) = E(x_{i2} u_i) = 0$, where the residual u_i is defined by $u_i = y_i - \beta_{01}x_{i1} - \beta_{02}x_{i2}$. We want to impose the restriction that $\beta_{02} = 0$. Suppose that we first obtain $(\hat{\beta}_{1n}, \hat{\beta}_{2n})$ from the unrestricted least squares regression of y on x_1, x_2 , and then obtain a restricted estimator of $\theta_0 = \beta_{01}$ by choosing $\hat{\theta}_n$ to

$$\min_{\theta} \left(\begin{bmatrix} \hat{\beta}_{1n} \\ \hat{\beta}_{2n} \end{bmatrix} - \begin{bmatrix} \theta \\ 0 \end{bmatrix} \right)' A_n \left(\begin{bmatrix} \hat{\beta}_{1n} \\ \hat{\beta}_{2n} \end{bmatrix} - \begin{bmatrix} \theta \\ 0 \end{bmatrix} \right), \quad (1.1)$$

where A_n converges with probability one to A , a positive-definite matrix. The optimal choice for A is the inverse of the asymptotic covariance matrix of $(\hat{\beta}_{1n}, \hat{\beta}_{2n})$.

*I am grateful to Arthur Goldberger, Charles Manski, and a referee for helpful discussions and comments. Financial support was provided by the National Science Foundation, the University of Wisconsin Graduate School, and an Alfred P. Sloan Research Fellowship.

In the homoskedastic case with $E(u_i^2|x_{i1}, x_{i2}) = \sigma^2$, we can set $A_n = n^{-1} \sum_{i=1}^n x_i x_i'$ [where $x_i' = (x_{i1}, x_{i2})$], and then $\hat{\theta}_n = b_{yx_1}$, the least squares coefficient in the regression of y on x_1 . More generally, though, we can allow for heteroskedasticity of unknown form, as in White (1980a, b), and let

$$A_n = \left[\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \hat{u}_{in}^2 x_i x_i' \right) \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \right]^{-1}, \quad (1.2)$$

where

$$\hat{u}_{in} = y_i - \hat{\beta}_{1n} x_{i1} - \hat{\beta}_{2n} x_{i2}.$$

We know from the theory of minimum- χ^2 or minimum-distance estimation that the limiting distribution of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ depends only upon A , and not upon the sequence of estimators $\{A_n\}$. So if $E(u_i^2 x_i x_i') = E(u_i^2) E(x_i x_i')$, then $\hat{\theta}_n$ has the same asymptotic variance as b_{yx_1} ; but in general it is strictly more efficient.

Efficiency gains of this sort have been considered by Chamberlain (1982), Cragg (1983), Hansen (1982), and White (1982). We want to determine the lower bound on asymptotic variance for a consistent estimator of θ_0 , when all that is known is that

$$E[x_{i1}(y_i - \theta_0 x_{i1})] = E[x_{i2}(y_i - \theta_0 x_{i1})] = 0. \quad (1.3)$$

More generally, we shall work with moment restrictions of the form considered by Hansen (1982):

$$E\psi(z_i, \theta_0) = 0, \quad (1.4)$$

where $\{z_i\}_{i=1}^\infty$ is an independent sequence of $m \times 1$ random vectors distributed according to F_0 , θ_0 is $p \times 1$, and ψ is a known function with q components ($q \geq p$). Hansen developed generalized method of moments (GMM) estimators of the following sort: choose $\hat{\theta}_n$ to

$$\min_{\theta} \left[\frac{1}{n} \sum_{i=1}^n \psi(z_i, \theta) \right]' A_n \left[\frac{1}{n} \sum_{i=1}^n \psi(z_i, \theta) \right], \quad (1.5)$$

where A_n converges with probability one to A , a positive-definite matrix.¹ Under suitable regularity conditions, the optimal choice for A is $[E\psi(z, \theta_0)\psi'(z, \theta_0)]^{-1}$, and then the limiting distribution of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ is

¹Huber (1967) considered moment equations of the form (1.4) with $q = p$. Also see Burguete, Gallant and Souza (1982). Manski (1983) developed a generalization that is based directly on the empirical distribution function.

normal with covariance matrix equal to

$$\Lambda_0 = \left(\left[E \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' [E \psi(z, \theta_0) \psi'(z, \theta_0)]^{-1} \left[E \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}. \quad (1.6)$$

This framework includes the non-linear instrumental variable estimators of Amemiya (1974, 1977) and Jorgenson and Laffont (1974), as well as extensions that allow for heteroskedasticity of unknown form. We want to show that Λ_0 is in fact the best one can do, if all that is known is that F_0 satisfies the moment restrictions in (1.4).

The simplest version of this problem has $m = p = q = 1$ and $\psi(z, \theta) = z - \theta$, so that we are estimating a mean. The GMM estimator is the sample mean, $\bar{z}_n = n^{-1} \sum_{i=1}^n z_i$, and Λ_0 reduces to the familiar $E(z - \theta_0)^2$. Can we do better? The question is of some difficulty due to the non-parametric aspect. If we specified a parametric form for F_0 , then standard results based on the information matrix would apply. If F_0 is normal, then $\bar{z}_n$ is efficient, but in general we can do better with a non-normal F_0 in the parametric case. Even if the form of F_0 is not specified up to a parameter vector of finite dimension, perhaps the form could be estimated sufficiently well to apply a maximum likelihood type estimator.

Suppose that z_i has a multinomial distribution: $\text{Prob}\{z_i = \phi_j\} = \pi_{0j}$ ($j = 1, \dots, r$). The ϕ_j are known; the unknown parameters are $\pi_{0j} = F_0\{\phi_j\}$. Define $d_{ij} = 1$ if $z_i = \phi_j$, $d_{ij} = 0$ otherwise. It is well-known that the maximum likelihood estimator of π_{0j} is the sample frequency: $\hat{\pi}_{jn} = n^{-1} \sum_{i=1}^n d_{ij}$. The maximum likelihood estimator of $\theta_0 = \sum_{j=1}^r \pi_{0j} \phi_j$ is

$$\hat{\theta}_n = \sum_{j=1}^r \hat{\pi}_{jn} \phi_j = \bar{z}_n. \quad (1.7)$$

Hence $\Lambda_0 = E(z - \theta_0)^2$ does provide a lower bound on asymptotic variance for the multinomial case. Since any distribution can be approximated arbitrarily well by a multinomial distribution, it is plausible that Λ_0 is the lower bound for general F_0 .

Our key result is that this multinomial point extends to a general ψ function. We assume that F_0 has finite support $\{\phi_1, \dots, \phi_r\}$, so that the moment restrictions in (1.4) become

$$\sum_{j=1}^r \pi_{0j} \psi(\phi_j, \theta_0) = 0. \quad (1.8)$$

Now we have a parametric problem, and we can work out the information matrix bound on the asymptotic variance of a consistent estimator of θ_0 . The result is Λ_0 . Then we show that any F_0 satisfying the restrictions in (1.4) can be approximated arbitrarily well by a multinomial distribution satisfying (1.4). From there it is not hard to show that Λ_0 is the bound for general F_0 .

The case of a finite set of moment restrictions has also been treated, using a very different approach, by Levit (1974, 1975) and Nevel'son (1977).

Next we derive the bound on asymptotic efficiency when there are conditional moment restrictions. These restrictions have the following form:

$$E[\rho(x, y, \theta_0)|x] = 0 \quad (1.9)$$

with probability one. If, for example, $\rho(x, y, \theta) = y - \theta x$, then the regression function is restricted to be linear: $E(y|x) = \theta_0 x$. This is, of course, much more restrictive than the linear predictor specification that $E[x(y - \theta_0 x)] = 0$. The conditional moment restrictions imply that $E[g(x)(y - \theta_0 x)] = 0$ for any function g ; hence there is an infinite set of moment restrictions. Amemiya (1983) and MaCurdy (1982) consider methods for imposing a subset of these restrictions.

Once again a multinomial approximation is very fruitful. In that case x takes on only a finite set of values, $\{\tau_1, \dots, \tau_l\}$, and so in fact the number of moment restrictions is finite. Define $h_j(x) = 1$ if $x = \tau_j$, $h_j(x) = 0$ otherwise. Then

$$\begin{aligned} E[h_j(x)\rho(x, y, \theta)] &= E\{h_j(x)E[\rho(x, y, \theta)|x]\} \\ &= \text{Prob}\{x = \tau_j\}E[\rho(x, y, \theta)|x = \tau_j]. \end{aligned} \quad (1.10)$$

So $E[\rho(x, y, \theta)|x] = 0$ with probability one if and only if $E[h_j(x)\rho(x, y, \theta)] = 0$ for $j = 1, \dots, l$. Hence we can construct a ψ function based upon $h_j(x)\rho(x, y, \theta)$ and apply the Λ_0 bound. Evaluating Λ_0 for this ψ function gives

$$\Lambda_0^* = \{E[D_0'(x)\Sigma_0^{-1}(x)D_0(x)]\}^{-1}, \quad (1.11)$$

where

$$D_0(x) = E\left[\frac{\partial \rho(x, y, \theta_0)}{\partial \theta'}|x\right],$$

and

$$\Sigma_0(x) = E[\rho(x, y, \theta_0)\rho'(x, y, \theta_0)|x].$$

We show that any F_0 satisfying the conditional moment restrictions in (1.9) can be approximated arbitrarily well by a multinomial distribution satisfying (1.9). Then it follows that Λ_0^* is the bound for a general F_0 .

To interpret this result, consider again the regression model, $E(y|x) = \theta_0 x$. If we knew the conditional variance function, $\sigma_0^2(x) = E[(y - \theta_0 x)^2|x]$, then we could apply the weighted least squares estimator that regresses $y/\sigma_0(x)$ on $x/\sigma_0(x)$. The asymptotic variance of that estimator is $[E(x^2/\sigma_0^2(x))]^{-1}$, which equals Λ_0^* .

We also show that there is a sequence of GMM estimators that can be used to come arbitrarily close to the Λ_0^* bound. In the regression example, we can set

$$\psi'_k(x, y, \theta) = [y - \theta x, x(y - \theta x), \dots, x^{k-1}(y - \theta x)], \quad (1.12)$$

so that $1, x, \dots, x^{k-1}$ are being used as instrumental variables. The asymptotic variance of the GMM estimator is Λ_{0k} , from (1.6), and $\Lambda_{0k} \rightarrow \Lambda_0^*$ as $k \rightarrow \infty$ if the support of x is bounded.

Amemiya (1977) considered a non-linear simultaneous equations model of the form

$$\rho(x_i, y_i, \theta_0) = u_i. \quad (1.13)$$

He derived the best non-linear three-stage least squares estimator for the homoskedastic case, and he noted that it is difficult to apply since it depends upon $E[\partial \rho(x, y, \theta_0)/\partial \theta' | x]$. He raised the question of whether there might be a better estimator which, like the instrumental variable estimators, has the desirable property of remaining consistent when the form of the u_i distribution is unknown. (He showed that the quasi-maximum likelihood estimator based on a normal distribution does not have this property.)

Our Λ_0^* corresponds to the asymptotic covariance matrix of Amemiya's best non-linear three-stage least squares estimator. The difference is that we allow for heteroskedasticity of unknown form. Our results imply that Λ_0^* can be approximately attained by a feasible GMM estimator. If one assumes only that $E(u_i|x_i) = 0$, then one cannot do better than Λ_0^* . There remains the case in which u_i is assumed to be independent of x_i . Some results for this case are contained in Manski (1984).²

²Also see MaCurdy (1982); he derives efficiency gains from exploiting moment restrictions of the form $E[g(x_i)(u_i' - E(u_i'))] = 0$.

2. Moment restrictions: The multinomial case

2.1. The information matrix

The observations on the i th unit are contained in a $m \times 1$ vector z_i ; we assume that $\{z_i\}_{i=1}^\infty$ is a sequence of independent and identically distributed (i.i.d.) random vectors with probability measure F_0 : $\text{Prob}\{z_i \in B\} = F_0(B)$ for any Borel set B in R^m . We define the support of a probability measure F to be the smallest closed set S with $F(S) = 1$. Throughout the paper, all probability measures over R^m will have supports contained in Z , a Borel subset of R^m .

We are given a function ψ which maps a subset of $Z \times R^p$ into R^q . The restrictions on F_0 take the following moment form:

$$E_{F_0} \psi(z, \theta_0) = \int \psi(z, \theta_0) F_0(dz) = 0,$$

where θ_0 is some point in R^p ($p \leq q$). More generally, we consider pairs (F, θ) that satisfy the following conditions:

Condition (C₁). (i) θ is in an open set $\Theta \subset R^p$ such that $\psi(z, \beta)$ and $\partial \psi(z, \beta) / \partial \beta'$ are continuous for $(z, \beta) \in Z \times \Theta$; (ii) $E_F \psi(z, \theta) = 0$; (iii) $E_F[\psi(z, \theta) \psi'(z, \theta)]$ is positive-definite; (iv) $\text{rank } E_F[\partial \psi(z, \theta) / \partial \theta'] = p$.

In this section we shall consider the case in which z_i has a multinomial distribution; i.e., the support of F_0 is a finite set $\{\phi_1, \dots, \phi_r\}$. The ϕ_j are known; the unknown parameters are $\pi_{0j} = F_0\{\phi_j\}$. These parameters are subject to the restriction that $\sum_{j=1}^r \pi_{0j} \psi(\phi_j, \theta_0) = 0$. Now we can apply standard results on efficient estimation in parametric models. Since the results for the multinomial case will be used again when we treat the case of a general F_0 , we shall distinguish the multinomial case by using G instead of F for the probability measures.

Lemma 1. Suppose that (G_0, θ_0) satisfies Condition (C₁) and that G_0 has finite support $\{\phi_1, \dots, \phi_r\}$. Then there is an open set $V \subset R^s$ ($s = r - (q - p) - 1$) and a family of probability measures $\{G_\gamma: \gamma \in V\}$ such that for all $\gamma \in V$: (i) G_γ has support $\{\phi_1, \dots, \phi_r\}$; (ii) (G_γ, γ^1) satisfies Condition (C₁), where γ^1 contains the first p elements of γ ; (iii) there is a $\gamma_0 \in V$ with $\gamma_0^1 = \theta_0$ and $G_{\gamma_0} = G_0$; (iv) $\pi_j(\gamma) \equiv G_\gamma\{\phi_j\}$ is a continuously differentiable function of γ ($j = 1, \dots, r$), and

$$J_\gamma = \sum_{j=1}^r \pi_j^{-1}(\gamma) \frac{\partial \pi_j(\gamma)}{\partial \gamma} \frac{\partial \pi_j(\gamma)}{\partial \gamma'}$$

is positive-definite. (Proof in appendix.)

Lemma 1 sets up a parametric family $\{G_\gamma\}$ of multinomial distributions. They all have known support $\{\phi_1, \dots, \phi_r\}$ and satisfy the moment restrictions

$$E_{G_\gamma} \psi(z, \gamma^1) = \sum_{j=1}^r \pi_j(\gamma) \psi(\phi_j, \gamma^1) = 0,$$

where $\pi_j(\gamma) = G_\gamma\{\phi_j\}$. Given independent observations $z_1, \dots, z_n$ distributed according to G_{γ_0} ($= G_0$), the problem is to estimate γ_0 and in particular θ_0 , the subvector of γ_0 containing the first p components.

Under suitable regularity conditions, the lower bound on the asymptotic variance of a consistent estimator of γ_0 is determined by the information matrix

$$\begin{aligned} J_0 &= \sum_{j=1}^r \pi_j(\gamma_0) \frac{\partial \log \pi_j(\gamma_0)}{\partial \gamma} \frac{\partial \log \pi_j(\gamma_0)}{\partial \gamma'} \\ &= \sum_{j=1}^r \pi_j^{-1}(\gamma_0) \frac{\partial \pi_j(\gamma_0)}{\partial \gamma} \frac{\partial \pi_j(\gamma_0)}{\partial \gamma'}. \end{aligned}$$

Define J_0^{11} to be the upper-left $p \times p$ block of J_0^{-1} . Then J_0^{11} provides the information matrix bound on the asymptotic variance of a consistent estimator of θ_0 . Our key result is contained in the following lemma:

Lemma 2. Under the assumptions of Lemma 1,

$$\begin{aligned} J_0^{11} &= \left(\left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \right. \\ &\quad \left. \times \left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}. \end{aligned}$$

(Proof in appendix.)

Lemma 2 asserts that knowing that the support of G_0 is a given finite set does not increase the asymptotic efficiency of our estimates. The lower bound on asymptotic variance is precisely the asymptotic variance that can be achieved in the general case by using the GMM estimator.

To provide some insight, we shall give a heuristic derivation of this result. Define $d_{ij} = 1$ if $z_i = \phi_j$, $d_{ij} = 0$ otherwise, and let $\hat{\pi}_{jn} = n^{-1} \sum_{i=1}^n d_{ij}$ ($j = 1, \dots, r$). Define $\pi'_0 = (\pi_{01}, \dots, \pi_{0,r-1})$ and $\hat{\pi}'_n = (\hat{\pi}_{1n}, \dots, \hat{\pi}_{r-1,n})$. The central limit theorem implies that the limiting distribution of $\sqrt{n}(\hat{\pi}_n - \pi_0)$ is $N(0, \Omega)$.

The generalized method of moments estimator chooses $\hat{\theta}_n$ to

$$\min_{\theta} s_n'(\theta) A_n s_n(\theta), \quad (2.1)$$

where

$$s_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi(z_i, \theta) = \sum_{j=1}^r \hat{\pi}_{jn} \psi(\phi_j, \theta),$$

and A_n converges with probability one to $[E_{G_0} \psi(z, \theta_0) \psi'(z, \theta_0)]^{-1}$.

Suppose that we augment the moment vector $s_n(\theta)$ as follows: define

$$\gamma = \begin{bmatrix} \theta \\ \tau \end{bmatrix}, \quad \gamma_0 = \begin{bmatrix} \theta_0 \\ \tau_0 \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \Phi \pi_0 \end{bmatrix},$$

and let

$$h(\hat{\pi}_n, \gamma) = \begin{bmatrix} \sum_{j=1}^r \hat{\pi}_{jn} \psi(\phi_j, \theta) \\ \Phi \hat{\pi}_n - \tau \end{bmatrix},$$

where the $(r-1-q) \times (r-1)$ matrix Φ is chosen so that $\partial h(\pi_0, \gamma_0) / \partial \pi'$ is non-singular. Then the restrictions on π_0 are given by $h(\pi_0, \gamma_0) = 0$. Note that the additional moment equations do not add any restrictions: $\Phi \pi_0 - \tau_0 = 0$ is simply the definition of τ_0 .

Since $\partial h(\pi_0, \gamma_0) / \partial \pi'$ is non-singular, we can use the implicit function theorem to construct a function $\pi'(\gamma) = (\pi_1(\gamma), \dots, \pi_{r-1}(\gamma))$ such that $h(\pi(\gamma), \gamma) = 0$ and $\pi(\gamma_0) = \pi_0$. Then

$$G_\gamma\{\phi_j\} = \pi_j(\gamma), \quad j = 1, \dots, r-1,$$

$$G_\gamma\{\phi_r\} = 1 - \sum_{j=1}^{r-1} \pi_j(\gamma)$$

defines a family $\{G_\gamma\}$ of probability measures that satisfies the restrictions. The data are generated according to G_{γ_0} ($= G_0$) and $\pi_0 = \pi(\gamma_0)$ expresses the $(q-p)$ restrictions in explicit form. [Note that π_0 is $(r-1) \times 1$ and γ_0 is $(r-1-(q-p)) \times 1$.] Furthermore,

$$\frac{\partial \pi(\gamma_0)}{\partial \gamma'} = - \left[\frac{\partial h(\pi_0, \gamma_0)}{\partial \pi'} \right]^{-1} \left[\frac{\partial h(\pi_0, \gamma_0)}{\partial \gamma'} \right].$$

Consider the following estimator: Choose $\hat{\gamma}'_n = (\hat{\theta}'_n, \hat{\tau}'_n)$ to

$$\min_{\gamma} h'(\hat{\pi}_n, \gamma) [\hat{V}_n(h)]^{-1} h(\hat{\pi}_n, \gamma), \quad (2.2)$$

where

$$\lim_{n \rightarrow \infty} \hat{V}_n(h) = \frac{\partial h(\pi_0, \gamma_0)}{\partial \pi'} \Omega \frac{\partial h'(\pi_0, \gamma_0)}{\partial \pi}$$

with probability one. Since the additional moment equations do not add any restrictions, the asymptotic variances of $\hat{\theta}_n$ in (2.1) and (2.2) are identical. In fact, if the upper-left $q \times q$ block of $\hat{V}_n(h)$ equals A_n^{-1} , then the $\hat{\theta}_n$ obtained from (2.1) and (2.2) are identical.³

The first-order conditions in (2.2) together with the mean-value theorem give

$$\begin{aligned} 0 &= \frac{\partial h'}{\partial \gamma} [\hat{V}_n(h)]^{-1} h(\hat{\pi}_n, \hat{\gamma}_n) \\ &= \frac{\partial h'}{\partial \gamma} \left(\frac{\partial h'}{\partial \pi} \right)^{-1} \hat{\Omega}_n^{-1} \left(\frac{\partial h}{\partial \pi'} \right)^{-1} \left[\frac{\partial h}{\partial \pi'} (\hat{\pi}_n - \pi_0) + \frac{\partial h}{\partial \gamma'} (\hat{\gamma}_n - \gamma_0) \right] \\ &= - \frac{\partial \pi'}{\partial \gamma} \hat{\Omega}_n^{-1} \left[\hat{\pi}_n - \pi_0 - \frac{\partial \pi}{\partial \gamma'} (\hat{\gamma}_n - \gamma_0) \right], \end{aligned}$$

where the partial derivatives are evaluated at the appropriate points.

Now consider imposing the explicit form of the restrictions: Choose $\hat{\gamma}_n$ to

$$\min_{\gamma} [\hat{\pi}_n - \pi(\gamma)]' \hat{\Omega}_n^{-1} [\hat{\pi}_n - \pi(\gamma)]. \quad (2.3)$$

The first-order conditions together with the mean-value theorem give

$$0 = \frac{\partial \pi'}{\partial \gamma} \hat{\Omega}_n^{-1} \left[\hat{\pi}_n - \pi_0 - \frac{\partial \pi}{\partial \gamma'} (\hat{\gamma}_n - \gamma_0) \right].$$

So in both (2.2) and (2.3), the limiting distribution of $\sqrt{n}(\hat{\gamma}_n - \gamma_0)$ coincides with the limiting distribution of

$$\left[\frac{\partial \pi'(\gamma_0)}{\partial \gamma} \Omega^{-1} \frac{\partial \pi(\gamma_0)}{\partial \gamma'} \right]^{-1} \frac{\partial \pi'(\gamma_0)}{\partial \gamma} \Omega^{-1} [\sqrt{n}(\hat{\pi}_n - \pi_0)];$$

in both cases the limiting distribution is

$$N\left(0, \left[\frac{\partial \pi'(\gamma_0)}{\partial \gamma} \Omega^{-1} \frac{\partial \pi(\gamma_0)}{\partial \gamma'} \right]^{-1}\right).$$

³Chamberlain (1982, proposition 9).

It is well known from the theory of minimum- χ^2 or minimum-distance estimation that the estimator in (2.3) is asymptotically equivalent to maximum likelihood.⁴ Hence the GMM estimator in (2.1) is asymptotically efficient for the multinomial case.

In the next section, we shall state the asymptotic efficiency result more carefully, using results on local asymptotic minimax estimation. These results will be used again for the general case in which F need not be multinomial.

2.2. Local asymptotic minimax bounds

There exist superefficient estimators that have limiting normal distributions with variance less than that of the maximum likelihood estimator at certain points in the parameter space; however, these estimators do poorly in neighborhoods of the points of superefficiency. [See Le Cam (1953).] So the information matrix provides a bound on asymptotic variance only if we restrict attention to sufficiently well-behaved estimators. A very general way of doing this is provided by the local asymptotic minimax criterion.

First we specify a class L of loss functions: $l \in L$ if for all $u, v \in R$,

- (i) $l(u) = l(|u|)$,
- (ii) $|u| \leq |v|$ implies $l(u) \leq l(v)$,
- (iii) $\int_{-\infty}^{\infty} l(u) \exp(-\frac{1}{2}\lambda u^2) du < \infty$ for $\lambda > 0$,
- (iv) $l(0) = 0$.

Let γ_1 be the first component of γ and let $T_n(z_1, \dots, z_n)$ be a (measurable) estimator of γ_1 based on the first n observations. If the z_i are i.i.d. according to F , then the expected loss, normalized by $\sqrt{n}$, is

$$E_F \{ l[\sqrt{n}(T_n - \gamma_1)] \} = \int l[\sqrt{n}(T_n(z_1, \dots, z_n) - \gamma_1)] \\ \times F^n(d(z_1, \dots, z_n)),$$

where F^n is the product measure:

$$F^n(B_1 \times \dots \times B_n) = \prod_{i=1}^n F(B_i).$$

⁴See Neyman (1949) and Ferguson (1958). In our notation, $\Omega = II - \pi_0 \pi_0'$, where $II = \text{diag}\{\pi_{01}, \dots, \pi_{0, r-1}\}$, and $1 - \pi_0' II^{-1} \pi_0 = \pi_{0r}$ implies that $\Omega^{-1} = II^{-1} + \pi_{0r}^{-1} \iota \iota'$, where ι is an $(r-1) \times 1$ vector of ones; hence $\iota' \partial \pi / \partial \gamma' = -\partial \pi_r / \partial \gamma'$ implies that $(\partial \pi'(\gamma_0) / \partial \gamma) \Omega^{-1} \times (\partial \pi(\gamma_0) / \partial \gamma') = \sum_{j=1}^r \pi_j^{-1}(\gamma_0) (\partial \pi_j(\gamma_0) / \partial \gamma) (\partial \pi_j(\gamma_0) / \partial \gamma')$, which is the information matrix.

We are given a parametric family $\{F_\gamma: \gamma \in V\}$, where V is an open subset of R^s ; the observations z_i are i.i.d. according to F_{γ_0} , where $\gamma_0 \in V$. In order to exclude ill-behaved estimators, we consider the maximum expected loss as γ varies over the small neighborhood of γ_0 given by $\|\gamma - \gamma_0\| < \delta$ ($\|\cdot\|$ is the standard Euclidean norm). A theorem of Hájek (1972) provides conditions under which

$$\lim_{\delta \rightarrow 0} \liminf_{n \rightarrow \infty} \sup_{\|\gamma - \gamma_0\| < \delta} E_{F_\gamma} \left\{ l \left[\sqrt{n} (T_n - \gamma_1) \right] \right\} \\ \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} l(\sigma_0 u) \exp\left(-\frac{1}{2}u^2\right) du,$$

with

$$\sigma_0 = (\sigma_{0,1,1})^{1/2} \quad \text{and} \quad \{\sigma_{0,j,k}\}_{j,k=1}^p = J_0^{-1},$$

where J_0 is the information matrix evaluated at γ_0 . The theorem asserts that for large n , the maximum expected loss in a small neighborhood of γ_0 is bounded below by the expected loss that would obtain if $\sqrt{n}(T_n - \gamma_1)$ had a normal distribution with variance given by the (1,1) element of J_0^{-1} .

The theorem assumes that $\{F_\gamma: \gamma \in V\}$ satisfies a local asymptotic normality (LAN) condition at γ_0 . The LAN condition requires that, for γ in a shrinking neighborhood of γ_0 , the log-likelihood ratio is approximately (as $n \rightarrow \infty$) normally distributed. Convenient sufficient conditions for LAN are given by Roussas (1972). It is straightforward to verify these conditions for the $\{G_\gamma: \gamma \in V\}$ family of Lemma 1; then Lemma 2 gives the following bound.

Theorem 1. Suppose that the assumptions of Lemma 1 hold. Let γ_1 be the first component of γ and let $T_n(z_1, \dots, z_n)$ be any (measurable) estimator of γ_1 . Then for any loss function $l \in L$, we have

$$\lim_{\delta \rightarrow 0} \liminf_{n \rightarrow \infty} \sup_{\|\gamma - \gamma_0\| < \delta} E_{G_\gamma} \left\{ l \left[\sqrt{n} (T_n - \gamma_1) \right] \right\} \\ \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} l(\sigma_0 u) \exp\left(-\frac{1}{2}u^2\right) du,$$

where $\sigma_0 = (\sigma_{0,1,1})^{1/2}$ and

$$\{\sigma_{0,j,k}\}_{j,k=1}^p = \left(\left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \right. \\ \left. \times \left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}$$

(Proof in appendix.)

We have established the lower bound on asymptotic risk for the multinomial case. Since data are measured in discrete units, one can argue that this result is all that is needed. Nevertheless, I would be uncomfortable if the restriction to a finite support were essential. In the next section, we shall use the multinomial result to treat the general case.

3. Moment restrictions: The general case

Now the observations z_i are assumed to be distributed according to a general probability measure F_0 , where (F_0, θ_0) satisfies Condition (C_1) . In order to obtain a local minimax bound, we need to define a neighborhood of F_0 .

Let Π be the set of all probability measures F defined on the Borel sets in R^m with $F(Z) = 1$. Given an $F \in \Pi$, we define the basic neighborhoods of F to be sets of the form

$$\left\{ Q \in \Pi: \left| \int f_j dQ - \int f_j dF \right| < \epsilon_j, j = 1, \dots, k \right\},$$

where $\epsilon_j > 0$, k is some positive integer, and $f_j: Z \rightarrow R$ are measurable functions such that $\int |f_j| dF < \infty$. An arbitrary neighborhood of F is formed by taking unions of sets of this form.

If we had restricted the f_j to be bounded and continuous, then our neighborhoods would correspond to the topology of weak convergence: $\int f dF_n \rightarrow \int f dF$ for all bounded continuous functions $f: Z \rightarrow R$ is equivalent to the convergence of the corresponding distribution functions at continuity points.⁵ But using weak neighborhoods would not give as strong a result in Theorem 2, since we shall be providing a lower bound on expected loss as F varies over an arbitrarily small neighborhood of F_0 . This might not matter if the distributions are assumed to have bounded support; but if Z is unbounded, then a weak neighborhood of F_0 contains distributions with infinite variance.

Theorem 2. Suppose that (F_0, θ_0) satisfies Condition (C_1) . Let Δ be any neighborhood of (F_0, θ_0) , and let Γ be the subset of Δ such that Condition (C_1) is satisfied for all $(F, \theta) \in \Gamma$. Let θ_1 be the first component of θ and let $T_n(z_1, \dots, z_n)$ be any (measurable) estimator of θ_1 . Then for any $l \in L$ we have

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \sup_{(F, \theta) \in \Gamma} E_F \left\{ l \left[\sqrt{n} (T_n - \theta_1) \right] \right\} \\ & \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} l(\sigma_0 u) \exp\left(-\frac{1}{2}u^2\right) du, \end{aligned}$$

⁵See Billingsley (1968, p. 236).

where $\sigma_0 = (\sigma_{0,1,1})^{1/2}$ and

$$\begin{aligned} \{\sigma_{0,j,k}\}_{j,k=1}^p &= \Lambda_0 \\ &= \left(\left[E_{F_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' \left[E_{F_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \right. \\ &\quad \left. \times \left[E_{F_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}. \end{aligned}$$

The theorem asserts that every neighborhood of F_0 contains a distribution F , which satisfies the restrictions, such that (for large n) the normalized risk in estimating θ_1 is bounded below by the risk for an estimator with a $N(\theta_1, \sigma_0^2)$ distribution, where σ_0^2 is the $(1,1)$ element of Λ_0 . The proof is based on setting up a multinomial distribution G_0 , where G_0 is arbitrarily close to F_0 ; (G_0, θ_0) satisfies (C_1) ; and

$$\int \psi(z, \theta_0) \psi'(z, \theta_0) G_0(dz) = \int \psi(z, \theta_0) \psi'(z, \theta_0) F_0(dz),$$

$$\int \frac{\partial \psi(z, \theta_0)}{\partial \theta'} G_0(dz) = \int \frac{\partial \psi(z, \theta_0)}{\partial \theta'} F_0(dz).$$

Then the bound on expected loss in Theorem 1 implies the bound in Theorem 2. The construction of G_0 is based on the following result:

Lemma 3. Let $h: Z \rightarrow R^l$ be a measurable function and let F be a probability measure with support $Z_F \subset Z$. If $\int \|h\| dF < \infty$, then there exists a probability measure G whose support is a finite subset of Z_F and $\int h dG = \int h dF$.

Proof. First we shall show that $\lambda = \int h dF$ is in K , the convex hull of $h(Z_F)$. Suppose it is not. Then the separating hyperplane theorem implies that there is a $\beta \in R^l$, $\beta \neq 0$, and $\eta \in R$ such that

$$\beta' \lambda - \eta = \int (\beta' h - \eta) dF \geq 0,$$

and

$$\beta' h(z) - \eta \leq 0 \quad \text{for all } z \in Z_F.$$

Hence with probability one $h(z)$ is on the hyperplane that separates λ and K : $F\{z \in Z: h(z) \in J_1\} = 1$, where $J_1 = \{x \in R^l: \beta' x = \eta\}$. Define $H_1 = h(Z_F) \cap J_1$ and let K_1 be the convex hull of H_1 relative to J_1 . Then $\lambda \in J_1$, and we can

separate λ from K_1 relative to J_1 . So as above we can show that with probability one $h(z)$ is in a $(t-2)$ -dimensional affine subspace. Continuing in this way, we obtain $h(z) = \lambda$ with probability one, contradicting $\lambda \notin K$; so $\lambda \in K$.

Hence λ lies in the convex hull of some subset of $h(Z_F)$ that contains at most $t+1$ points [Berge (1963, theorem 4, p. 147)]:

$$\lambda = \sum_{j=1}^r \alpha_j h(\phi_j), \quad r \leq t+1,$$

where $\alpha_j > 0$, $\sum_{j=1}^r \alpha_j = 1$, and $S = \{\phi_1, \dots, \phi_r\}$ is a subset of Z_F . Now define a probability measure G with support S by setting $G\{\phi_j\} = \alpha_j$ ($j = 1, \dots, r$). Then

$$\int h \, dG = \sum_{j=1}^r \alpha_j h(\phi_j) = \int h \, dF. \quad \text{Q.E.D.}$$

Proof of Theorem 2. From the definition of a neighborhood of F_0 , we know that there is an $\varepsilon > 0$ and measurable functions $f_j: Z \rightarrow R$ such that $(F, \theta) \in \Delta$ if $|\int f_j \, dF - \int f_j \, dF_0| < \varepsilon$ ($j = 1, \dots, k$) and $\|\theta - \theta_0\| < \varepsilon$. Let $h: Z \rightarrow R^t$ be the measurable function whose components consist of $f_1, \dots, f_k$, and the components of $\psi(\cdot, \theta_0)$, $\partial\psi(\cdot, \theta_0)/\partial\theta'$, and $\psi(\cdot, \theta_0)\psi'(\cdot, \theta_0)$. Then Lemma 3 asserts that there is a probability measure G_0 , whose support is a finite subset of Z , such that $\int h \, dG_0 = \int h \, dF_0$. Hence $(G_0, \theta_0) \in \Gamma$.

Now use Lemma 1 to construct the family of probability measures $\{G_\gamma\}$ such that (G_γ, γ^1) satisfies Condition (C_1) . Then for any function $f: Z \rightarrow R$,

$$\int f \, dG_\gamma = \sum_{j=1}^r \pi_j(\gamma) f(\phi_j) \rightarrow \sum_{j=1}^r \pi_j(\gamma_0) f(\phi_j) = \int f \, dG_0,$$

as $\gamma \rightarrow \gamma_0$. So there is a $\delta > 0$ such that $(G_\gamma, \gamma^1) \in \Gamma$ for $\|\gamma - \gamma_0\| < \delta$. Since

$$\sup_{(F, \theta) \in \Gamma} E_F \{I[\sqrt{n}(T_n - \theta_1)]\} \geq \sup_{\|\gamma - \gamma_0\| < \delta} E_{G_\gamma} \{I[\sqrt{n}(T_n - \gamma_1)]\},$$

we can apply Theorem 1 to obtain

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \sup_{(F, \theta) \in \Gamma} E_F \{I[\sqrt{n}(T_n - \theta_1)]\} \\ & \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} I(\sigma_0 u) \exp(-\tfrac{1}{2}u^2) \, du, \end{aligned}$$

where $\sigma_0 = (\sigma_{0,1,1})^{1/2}$ and

$$\begin{aligned} \{\sigma_{0,j,k}\}_{j,k=1}^p &= \left(\left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \right. \\ &\quad \left. \times \left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1} \\ &= \Lambda_0. \quad \text{Q.E.D.} \end{aligned}$$

4. Conditional moment restrictions

4.1. The minimax bound

We shall assume that $Z = X \times Y$, where X is a Borel subset of R^s and Y is a Borel subset of R^t ($s + t = m$). A probability measure F defined on the Borel sets of R^m can be represented by the product of a conditional probability measure $F(dy|x)$ and a marginal probability measure $F^*(dx)$; that is, if $g: R^s \times R^t \rightarrow R$ is an integrable function, then

$$\int g \, dF = \int \left[\int g(x, y) F(dy|x) \right] F^*(dx).$$

For each $x \in R^s$, $F(\cdot|x)$ is a probability measure on the Borel sets in R^t , and for each Borel set B , $F(B|\cdot)$ is a measurable function mapping R^s into R ; F^* is a probability measure on the Borel sets in R^s . A version of the conditional expectation is provided by

$$E_F[g(z)|x] = \int g(x, y) F(dy|x).$$

Let X_F be the support of F^* . We shall assume that $X_F \subset X$ and that the support of $F(dy|x)$ is contained in Y for every $x \in X_F$.

We are given a function ρ which maps a subset of $Z \times R^p$ into R^r . The observations $z_i = (x_i, y_i)$ are i.i.d. according to F_0 . Now the restrictions on F_0 involve conditional moments:

$$E_{F_0}[\rho(z, \theta_0)|x] = \int \rho(x, y, \theta_0) F_0(dy|x) = 0 \quad \text{for all } x \in X_{F_0},$$

where θ_0 is some point in R^p . More generally, we consider pairs (F, θ) that

satisfy the following condition:

Condition (C₂). (i) θ is in an open set $\Theta \subset R^p$ such that $\rho(z, \beta)$ and $\partial\rho(z, \beta)/\partial\beta'$ are continuous on $Z \times \Theta$; (ii) $E_F[\rho(z, \theta)|x] = 0$ for all $x \in X_F$; (iii) $\Sigma(x) \equiv E_F[\rho(z, \theta)\rho'(z, \theta)|x]$ is positive-definite for all $x \in X_F$; (iv) define $D(x) = E_F[\partial\rho(z, \theta)/\partial\theta'|x]$ for $x \in X_F$; then $E_{F_1}[D'(x)\Sigma^{-1}(x)D(x)]$ is positive-definite.

If (F_0, θ_0) satisfies Condition (C₂), then we shall use the following notation:

$$X_0 = X_{F_0},$$

$$\Sigma_0(x) = E_{F_0}[\rho(z, \theta_0)\rho'(z, \theta_0)|x],$$

$$D_0(x) = E_{F_0}[\partial\rho(z, \theta_0)/\partial\theta'|x],$$

$$\Lambda_0^* = \{E_{F_0}[D_0'(x)\Sigma_0^{-1}(x)D_0(x)]\}^{-1}.$$

Our proof of the minimax bound uses a multinomial distribution G_0 that has certain conditional moments in common with F_0 . The following lemma sets up G_0 .

Lemma 4. Suppose that (F_0, θ_0) satisfies Condition (C₂). Let $\mathcal{T}$ be any neighborhood of F_0 . Then there exists a probability measure $G_0 \in \mathcal{T}$ with finite support such that $X_{G_0} \subset X_0$ and for all $x \in X_{G_0}$,

$$E_{G_0}[\rho(z, \theta_0)|x] = 0,$$

$$E_{G_0}[\rho(z, \theta_0)\rho'(z, \theta_0)|x] = \Sigma_0(x),$$

$$E_{G_0}[\partial\rho(z, \theta_0)/\partial\theta'|x] = D_0(x).$$

Furthermore,

$$E_{G_0}[D_0'(x)\Sigma_0^{-1}(x)D_0(x)] = \Lambda_0^{*-1}.$$

(Proof in appendix.)

The proof of this result uses Lemma 3 to set up a multinomial G_0^* that is close to F_0^* and has

$$\int (D_0'\Sigma_0^{-1}D_0) dG_0^* = \int (D_0'\Sigma_0^{-1}D_0) dF_0^*.$$

Then at each point x in the finite support of G_0^* , we use Lemma 3 again to set up a multinomial $G_0(dy|x)$ that is close to $F_0(dy|x)$ and matches the $F_0(dy|x)$ moments for ρ , $\rho\rho'$, and $\partial\rho/\partial\theta'$.

Now we shall derive the minimax bound for the multinomial case. Let $\{\tau_1, \dots, \tau_l\} \subset X_0$ be the support of G_0^* . Let $h_j: X \rightarrow R$ be any continuous function that satisfies $h_j(\tau_j) = 1$, $h_j(\tau_k) = 0$ ($j, k = 1, \dots, l$; $j \neq k$). Define

$$B_0(x) = \begin{bmatrix} h_1(x)I_r \\ \vdots \\ h_l(x)I_r \end{bmatrix},$$

where I_r is the $r \times r$ identity matrix. Define $\psi_0: Z \times \Theta \rightarrow R^{lr}$ by

$$\psi_0(z, \theta) = B_0(x)\rho(x, y, \theta). \quad (4.1)$$

Then we have

$$\begin{aligned} E_{G_0}\psi_0(z, \theta) &= \int B_0(x) \left[\int \rho(x, y, \theta) G_0(dy|x) \right] G_0^*(dx) \\ &= \begin{bmatrix} G_0^*\{\tau_1\} E_{G_0}[\rho(z, \theta)|\tau_1] \\ \vdots \\ G_0^*\{\tau_l\} E_{G_0}[\rho(z, \theta)|\tau_l] \end{bmatrix}. \end{aligned}$$

Hence $E_{G_0}\psi_0(z, \theta) = 0$ if and only if $E_{G_0}[\rho(z, \theta)|x] = 0$ for every x in X_{G_0} — conditional moment restrictions on a multinomial distribution reduce to a finite set of moment restrictions.

So the bound on asymptotic variance is given by our results in section 2. We only need to evaluate

$$\left(\left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi_0(z, \theta_0) \psi_0'(z, \theta_0) \right]^{-1} \left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}.$$

First note that $E_{G_0}(\psi_0\psi_0')$ is the following block-diagonal matrix:

$$\begin{aligned} E_{G_0}[\psi_0(z, \theta_0) \psi_0'(z, \theta_0)] &= \sum_{j=1}^l G_0^*\{\tau_j\} B_0(\tau_j) \Sigma_0(\tau_j) B_0'(\tau_j) \\ &= \text{diag}\{G_0^*\{\tau_1\} \Sigma_0(\tau_1), \dots, G_0^*\{\tau_l\} \Sigma_0(\tau_l)\}, \end{aligned} \quad (4.2)$$

and

$$E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} = \sum_{j=1}^I G_0^* \{ \tau_j \} B_0(\tau_j) D_0(\tau_j) = \begin{bmatrix} G_0^* \{ \tau_1 \} D_0(\tau_1) \\ \vdots \\ G_0^* \{ \tau_I \} D_0(\tau_I) \end{bmatrix}.$$

Hence

$$\begin{aligned} & \left(\left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right]' [E_{G_0} \psi_0(z, \theta_0) \psi_0'(z, \theta_0)]^{-1} \left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right] \right)^{-1} \\ &= \left[\sum_{j=1}^I G_0^* \{ \tau_j \} D_0'(\tau_j) \Sigma_0^{-1}(\tau_j) D_0(\tau_j) \right]^{-1} \\ &= [E_{G_0} D_0'(x) \Sigma_0^{-1}(x) D_0(x)]^{-1} = \Lambda_0^*. \end{aligned} \quad (4.3)$$

We have shown that Λ_0^* is the bound on asymptotic variance when we have conditional moment restrictions in the multinomial case.

Since we can construct such a multinomial G_0 in any neighborhood of F_0 , we can show that Λ_0^* is the minimax bound in the general case.

Theorem 3. Suppose that (F_0, θ_0) satisfies Condition (C_2) . Let Δ be any neighborhood of (F_0, θ_0) and let Γ be the subset of Δ such that Condition (C_2) is satisfied for all $(F, \theta) \in \Gamma$. Let θ_1 be the first component of θ and let $T_n(z_1, \dots, z_n)$ by any (measurable) estimator of θ_1 . Then for any loss function $l \in L$, we have

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \sup_{(F, \theta) \in \Gamma} E_F \{ l[\sqrt{n}(T_n - \theta_1)] \} \\ & \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} l(\sigma_0 u) \exp(-\frac{1}{2}u^2) du, \end{aligned}$$

where

$$\sigma_0 = (\sigma_{0,1,1})^{1/2} \quad \text{and} \quad \{ \sigma_{0,j,k} \}_{j,k=1}^p = \Lambda_0^*.$$

Proof. Use Lemma 4 to construct G_0 such that $(G_0, \theta_0) \in \Gamma$. Define ψ_0 as in (4.1). It is straightforward to check that (G_0, θ_0) satisfies Condition (C_1) with $\psi = \psi_0$: $E_{G_0} \psi_0(z, \theta_0) = 0$; $E_{G_0} [\psi_0(z, \theta_0) \psi_0'(z, \theta_0)]$ is positive-definite by (4.2); $\text{rank } E_{G_0} [\partial \psi_0(z, \theta_0) / \partial \theta'] = p$ by (4.3). Hence we can use Lemma 1 to construct the family of probability measures $\{G_\gamma: \gamma \in V\}$ such that (G_γ, γ^1) satisfies Condition (C_1) . The support of G_γ coincides with that of G_0 . Hence the support of G_γ^* is $X_{G_\gamma} = \{\tau_1, \dots, \tau_I\}$, and $E_{G_\gamma} \psi_0(z, \gamma^1) = 0$ implies that

$$E_{G_\gamma} [\rho(z, \gamma^1) | x] = 0 \quad \text{for all } x \in X_{G_\gamma}.$$

Also (iii) and (iv) of Condition (C₁) imply that (iii) and (iv) of Condition (C₂) hold for (G_γ, γ^1) . Hence we can choose $\delta > 0$ such that $(G_\gamma, \gamma^1) \in \Gamma$ if $\|\gamma - \gamma_0\| < \delta$. Let γ_1 be the first component of γ . Then apply Theorem 1 to obtain

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \sup_{(F, \theta) \in \Gamma} E_F \{ l[\sqrt{n}(T_n - \theta_1)] \} \\ & \geq \liminf_{n \rightarrow \infty} \sup_{\|\gamma - \gamma_0\| < \delta} E_{G_\gamma} \{ l[\sqrt{n}(T_n - \gamma_1)] \} \\ & \geq (2\pi)^{-1/2} \int_{-\infty}^{\infty} l(\sigma_0 u) \exp(-\tfrac{1}{2}u^2) du, \end{aligned}$$

where $\sigma_0 = (\sigma_{0,1,1})^{1/2}$ and [by (4.3)]

$$\begin{aligned} \{ \sigma_{0,j,k} \}_{j,k=1}^p &= \left(\left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi_0(z, \theta_0) \psi'_0(z, \theta_0) \right]^{-1} \right. \\ & \quad \left. \times \left[E_{G_0} \frac{\partial \psi_0(z, \theta_0)}{\partial \theta'} \right] \right)^{-1} \\ &= \Lambda_0^*. \quad \text{Q.E.D.} \end{aligned}$$

There is a multidimensional version of Hájek's theorem in Ibragimov and Khas'minskii (1975, theorem 5.1; 1981, theorem 12.1, p. 162). They use the class of loss functions L' , which consists of the non-negative functions $l: R^p \rightarrow R$ with the following properties: (i) $l(u) = l(-u)$; (ii) $l(0) = 0$ and $l(u)$ is continuous at $u = 0$; (iii) the set $\{u: l(u) < \xi\}$ is convex for any $\xi > 0$; (iv) $\int_{R^p} l(u) \exp(-\lambda \|u\|^2) du < \infty$ for $\lambda > 0$. If we apply their results to the multinomial case, then we obtain the following version of the result in Theorem 3:

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \sup_{(F, \theta) \in \Gamma} E_F \{ l[\sqrt{n} \Lambda_0^{*-1/2}(T_n - \theta)] \} \\ & \geq (2\pi)^{-p/2} \int_{R^p} l(u) \exp(-\tfrac{1}{2}\|u\|^2) du, \end{aligned}$$

where l is any loss function in L' and $T_n(z_1, \dots, z_n)$ is any (measurable) estimator of θ .

4.2. Approximating the bound

Conditional moment restrictions are equivalent to a finite set of ordinary moment restrictions in the multinomial case. In the general case, conditional moment restrictions imply an infinite set of moment restrictions. Suppose that we form a finite subset of these restrictions as follows: let B be a continuous function mapping X into $q \times r$ matrices of real numbers. Define $\psi(z, \theta) = B(x)\rho(z, \theta)$. Then $E_{F_0}[\rho(z, \theta_0)|x] = 0$ implies that $E_{F_0}\psi(z, \theta_0) = 0$. Let Λ_0 be the asymptotic covariance matrix of the GMM estimator based on this ψ . Since Λ_0^* is the bound on the asymptotic covariance matrix when we impose all of the conditional moment restrictions, it ought to be true that $\Lambda_0 - \Lambda_0^*$ is positive semi-definite. We shall give a separate proof of this fact.

Proposition 1. Suppose that (F_0, θ_0) satisfies Condition (C_2) and let B be a continuous function mapping X into $q \times r$ matrices of real numbers. Define $\psi(z, \theta) = B(x)\rho(x, y, \theta)$ and assume that $E_{F_0}[\psi(z, \theta_0)\psi'(z, \theta_0)]$ is positive-definite and that $\text{rank } E_{F_0}[\partial\psi(z, \theta_0)/\partial\theta'] = p$. Then

$$\left(\left[E_{F_0} \frac{\partial\psi(z, \theta_0)}{\partial\theta'} \right]' \left[E_{F_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \left[E_{F_0} \frac{\partial\psi(z, \theta_0)}{\partial\theta'} \right] \right)^{-1} - \Lambda_0^*$$

is positive semi-definite.

Proof. Let α be an arbitrary point in R^p and define $w(x) = D_0(x)\alpha$. Note that

$$\alpha' \Lambda_0^{*-1} \alpha = E_{F_0} [w'(x) \Sigma_0^{-1}(x) w(x)].$$

It will suffice to show that

$$E_{F_0}(w' \Sigma_0^{-1} w) - E_{F_0}(w' B') [E_{F_0}(B \Sigma_0 B')]^{-1} E_{F_0}(B w) \geq 0. \quad (4.4)$$

Define

$$\beta = [E_{F_0}(B \Sigma_0 B')]^{-1} E_{F_0}(B w).$$

Then

$$E_{F_0} [(w - \Sigma_0 B \beta)' \Sigma_0^{-1} (w - \Sigma_0 B \beta)] \geq 0, \quad (4.5)$$

and a simple calculation shows that (4.5) is equivalent to (4.4). Q.E.D.

The idea of the proof comes from considering the space of random vectors with inner product $\langle v_1, v_2 \rangle = E_{F_0}(v_1' \Sigma_0^{-1} v_2)$. Then $\Sigma_0 B \beta$ is the orthogonal

projection of w onto the column space of $\Sigma_0 B'$, $v = w - \Sigma_0 B' \beta$ is the residual, and $\langle v, v \rangle \geq 0$ is equivalent to (4.4). This suggests that we can form a sequence $\{B_k\}$ that is sufficiently 'complete' so that w is approximately in the column space of $\Sigma_0 B'_k$ as $k \rightarrow \infty$. Then we shall have $v \rightarrow 0$, which will imply

$$\alpha' [\Lambda_0^{*-1} - \Lambda_{0k}^{-1}] \alpha \rightarrow 0,$$

where Λ_{0k} is the asymptotic covariance matrix of the GMM estimator based on $\psi_k = B_k \rho$. Since α is an arbitrary vector in R^p , this will imply $\Lambda_{0k} \rightarrow \Lambda_0^*$.

We need an appropriate definition of 'complete'. A measurable function $g: X \rightarrow R$ is in $L_2(F_0^*)$ if $\int g^2(x) F_0^*(dx) < \infty$. We shall say that a sequence $\{g_j\}_{j=1}^\infty$ of such functions is complete if for any $\varepsilon > 0$ and any function $h \in L_2(F_0^*)$, there are real numbers $\alpha_1, \dots, \alpha_q$ such that

$$\left\| h - \sum_{j=1}^q \alpha_j g_j \right\|_2 = \left\{ \int \left[h(x) - \sum_{j=1}^q \alpha_j g_j(x) \right]^2 F_0^*(dx) \right\}^{1/2} < \varepsilon.$$

Let $c_k(x) = 1$ if $\|x\| \leq k$ and $c_k(x) = 0$ if $\|x\| > k$. Since the continuous functions with bounded support are dense in $L_2(F_0^*)$ [Rudin (1974, theorem 3.14)], the Stone-Weierstrass theorem implies that we can let $g_j(x)$ have the form $c_{k_j}(x) \prod_{l=1}^s x_l^{k_{lj}}$, where k_j and k_{lj} are non-negative integers and $x = (x_1, \dots, x_s)$.

Suppose that $\{g_j\}$ is complete. If $\|g_k - \sum_{j=1}^{k-1} \alpha_j g_j\|_2 = 0$, then we can omit g_k and still have a complete sequence. Omitting in turn each such g_k gives a sequence $\{f_j\}$ which is linearly independent and complete. We shall say that $\{\Lambda_{0k}\}$ is based on a linearly independent, complete sequence $\{f_j\}$ if it is constructed as follows: define $B_k(x) = I_r \otimes \tilde{f}_k(x)$, where $\tilde{f}_k(x)$ is the $k \times 1$ vector whose j th component is $f_j(x)$.⁶ Define $\psi_k(z, \theta) = B_k(x) \rho(z, \theta)$. Then let

$$\begin{aligned} \Lambda_{0k} = & \left\{ \left[E_{F_0} \frac{\partial \psi_k(z, \theta_0)}{\partial \theta'} \right]' \left[E_{F_0} \psi_k(z, \theta_0) \psi_k'(z, \theta_0) \right]^{-1} \right. \\ & \left. \times \left[E_{F_0} \frac{\partial \psi_k(z, \theta_0)}{\partial \theta'} \right] \right\}^{-1}. \end{aligned}$$

We shall see that the relevant matrices are non-singular, at least for k sufficiently large. If F_0^* has finite support, then the $\{f_j\}$ sequence will be

⁶We use $\otimes$ to denote the Kronecker product.

finite: $\{f_1, \dots, f_l\}$. In that case we define $\Lambda_{0k} = \Lambda_{0l}$ for $k > l$. Then we have the following result:

Proposition 2. Suppose that (F_0, θ_0) satisfies Condition (C_2) , that $E_{F_0}[(\partial \rho(z, \theta_0)/\partial \theta')(\partial \rho'(z, \theta_0)/\partial \theta)] < \infty$, and that there is a $\zeta \in R$ such that $\zeta^{-1} \leq \alpha' \Sigma_0(x) \alpha \leq \zeta$ for all $x \in X_0$ and all $\alpha \in R^r$ with $\|\alpha\| = 1$. If $\{\Lambda_{0k}\}$ is based on a linearly independent, complete sequence $\{f_j\}$, then

$$\lim_{k \rightarrow \infty} \Lambda_{0k} = \Lambda_0^*.$$

(Proof in appendix.)

For a simple example, suppose that $m = 2$ and $E_{F_0}[(y - \theta_0 x)|x] = 0$. Then we can set

$$\psi'_k(z, \theta) = [y - \theta x, x(y - \theta x), \dots, x^{k-1}(y - \theta x)].$$

Λ_{0k} is the asymptotic variance of the GMM estimator that imposes optimally the restrictions in $E_{F_0} \psi_k(z, \theta_0) = 0$. Then if X_0 is bounded,

$$\lim_{k \rightarrow \infty} \Lambda_{0k} = \Lambda_0^* = \left[E_{F_0} \frac{x^2}{\sigma_0^2(x)} \right]^{-1},$$

where

$$\sigma_0^2(x) = E_{F_0}[(y - \theta_0 x)^2|x].$$

Note that we have not specified how k should increase as a function of sample size (and perhaps of the data) in order for the sequence of GMM estimators to have a limiting normal distribution with Λ_0^* as the covariance matrix; i.e., if $\hat{\theta}_n$ is a GMM estimator based on $\psi_k = B_k \rho$, then a result of interest that is distinct from ours is a specification for k as a function of $(n, z_1, \dots, z_n)$ such that $\sqrt{n}(\hat{\theta}_n - \theta_0)$ has a limiting $N(0, \Lambda_0^*)$ distribution as $n \rightarrow \infty$. Results with a flavor similar to Proposition 2 are in Amemiya (1973, theorem 5), Sims (1974, theorems 1, 2), and Hayashi and Sims (1983, theorem 5).

Appendix

Proof of Lemma 1. Define $\pi_0 \in R^r$ to have $\pi_{0j} = G_0\{\phi_j\}$ as the j th component. By assumption $\sum_{j=1}^r \pi_{0j} \psi(\phi_j, \theta_0) \psi'(\phi_j, \theta_0)$ is positive-definite. Hence the $q \times r$ matrix

$$\Psi = [\psi(\phi_1, \theta_0), \dots, \psi(\phi_r, \theta_0)]$$

has rank q . Define ι to be a $r \times 1$ vector of ones; ι cannot be in the row space of Ψ , for otherwise there is a $q \times 1$ vector α such that $\alpha' \Psi = \iota'$ and $\alpha' \Psi \pi_0 = \iota' \pi_0 = 1$; but this contradicts $\Psi \pi_0 = E_{G_0} \psi(z, \theta_0) = 0$. Hence there exists a $(r - q - 1) \times r$ matrix Φ such that the $r \times r$ matrix

$$\Delta = \begin{bmatrix} \Psi \\ \iota' \\ \Phi \end{bmatrix}$$

is non-singular.

Define $\tau_0 = \Phi \pi_0$ and $\gamma'_0 = (\theta'_0, \tau'_0)$. Let $A \subset R^r$ be a neighborhood of π_0 and let $B \subset R^p$ be a neighborhood of θ_0 . Define $h_1: A \times B \rightarrow R^q$ by $h_1(\pi, \theta) = \sum_{j=1}^r \pi_j \psi(\phi_j, \theta)$. Let $C \subset R^{r-q-1}$ be a neighborhood of τ_0 and define $h_2: A \times C \rightarrow R^{r-q-1}$ by $h_2(\pi, \tau) = \Phi \pi - \tau$. Let $\gamma' = (\theta', \tau')$ and define $h: A \times (B \times C) \rightarrow R^r$ by

$$h(\pi, \gamma) = \begin{bmatrix} h_1(\pi, \theta) \\ \iota' \pi - 1 \\ h_2(\pi, \tau) \end{bmatrix}.$$

We can choose B so that h is continuously differentiable with

$$\frac{\partial h(\pi, \gamma)}{\partial \pi'} = \begin{bmatrix} \psi(\phi_1, \theta), \dots, \psi(\phi_r, \theta) \\ \iota' \\ \Phi \end{bmatrix},$$

and

$$\frac{\partial h(\pi, \gamma)}{\partial \gamma'} = \begin{bmatrix} \sum_{j=1}^r \pi_j \frac{\partial \psi(\phi_j, \theta)}{\partial \theta'} & 0 \\ 0 & \dots & 0 \\ 0 & & -I \end{bmatrix}.$$

Note that $h(\pi_0, \gamma_0) = 0$ and $\partial h(\pi_0, \gamma_0) / \partial \pi' = \Delta$ is non-singular. Then it follows from the implicit function theorem that there exists a neighborhood V of γ_0 and a continuously differentiable function $f: V \rightarrow R^r$ such that $f(\gamma_0) = \pi_0$ and $h[f(\gamma), \gamma] = 0$ for $\gamma \in V$ [Loomis and Sternberg (1968, theorem 11.2)]. Furthermore, we have

$$\begin{aligned} \frac{\partial f(\gamma_0)}{\partial \gamma'} &= - \left[\frac{\partial h(\pi_0, \gamma_0)}{\partial \pi'} \right]^{-1} \frac{\partial h(\pi_0, \gamma_0)}{\partial \gamma'} \\ &= -\Delta^{-1} \begin{bmatrix} \Gamma & 0 \\ 0 & -I \end{bmatrix}, \end{aligned}$$

where

$$\Gamma = \begin{bmatrix} E_{G_0} & \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \\ 0 & \dots & 0 \end{bmatrix}$$

[Loomis and Sternberg (1968, theorem 11.1)]. Let Π_0 be the $r \times r$ diagonal matrix with π_{0j} as the j th diagonal element. Note that

$$\begin{aligned} J_{\gamma_0} &= \left[\frac{\partial f(\gamma_0)}{\partial \gamma'} \right]' \Pi_0^{-1} \frac{\partial f(\gamma_0)}{\partial \gamma'} \\ &= \begin{bmatrix} \Gamma' & 0 \\ 0 & -I \end{bmatrix} (\Delta \Pi_0 \Delta')^{-1} \begin{bmatrix} \Gamma & 0 \\ 0 & -I \end{bmatrix} \end{aligned} \quad (\text{A.1})$$

is positive-definite, since Γ has full column rank.

Let f_j be the j th component of f . We can choose V so that $f_j(\gamma) > 0$ for $\gamma \in V$ ($j = 1, \dots, r$). Since $\iota' f(\gamma) = 1$, we obtain a probability measure G_γ by setting $G_\gamma\{\phi_j\} = f_j(\gamma)$. We can also choose V so that (for all $\gamma \in V$) $E_{G_\gamma}[\psi(z, \theta)\psi'(z, \theta)]$ is positive-definite, $\text{rank } E_{G_\gamma}[\partial \psi(z, \theta)/\partial \theta'] = p$, and J_γ is positive-definite, since these conditions hold at $\gamma = \gamma_0$. Q.E.D.

Proof of Lemma 2. By (A.1),

$$J_0 = \begin{bmatrix} \Gamma' & 0 \\ 0 & -I \end{bmatrix} (\Delta \Pi_0 \Delta')^{-1} \begin{bmatrix} \Gamma & 0 \\ 0 & -I \end{bmatrix}.$$

Let $(\Delta \Pi_0 \Delta')_{11}$ denote the upper-left $(q+1) \times (q+1)$ block of $\Delta \Pi_0 \Delta'$. Then we have

$$(\Delta \Pi_0 \Delta')_{11} = \begin{bmatrix} \Psi \\ \iota' \end{bmatrix} \Pi_0 \begin{bmatrix} \Psi' & \iota \end{bmatrix} = \begin{bmatrix} \Psi \Pi_0 \Psi' & 0 \\ 0' & 1 \end{bmatrix},$$

since

$$\Psi \Pi_0 \iota = \Psi \pi_0 = E_{G_0} \psi(z, \theta_0) = 0.$$

Note that

$$\Psi \Pi_0 \Psi' = E_{G_0} [\psi(z, \theta_0) \psi'(z, \theta_0)].$$

Then applying Lemma A.1 gives

$$\begin{aligned} J_0^{11} &= \left[\Gamma' [(\Delta \Pi_0 \Delta')_{11}]^{-1} \Gamma \right]^{-1} \\ &= \left(\left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right]' \left[E_{G_0} \psi(z, \theta_0) \psi'(z, \theta_0) \right]^{-1} \right. \\ &\quad \left. \times \left[E_{G_0} \frac{\partial \psi(z, \theta_0)}{\partial \theta'} \right] \right)^{-1}. \quad \text{Q.E.D.} \end{aligned}$$

Lemma A.1. Define

$$\Sigma = \begin{bmatrix} \Gamma_1 & 0 \\ 0 & \Gamma_2 \end{bmatrix}' \Omega^{-1} \begin{bmatrix} \Gamma_1 & 0 \\ 0 & \Gamma_2 \end{bmatrix},$$

where Γ_1 is $(q+1) \times p$ and has rank p , Γ_2 is non-singular, and Ω is positive-definite. Let Σ^{11} be the upper-left $p \times p$ block of Σ^{-1} , and let Ω_{11} be the upper-left $(q+1) \times (q+1)$ block of Ω . Then

$$\Sigma^{11} = (\Gamma_1' \Omega_{11}^{-1} \Gamma_1)^{-1}.$$

Proof. Partition Σ and Ω^{-1} into (Σ_{jk}) and (Ω^{jk}) ($j, k = 1, 2$). Evaluating the partitioned inverse gives

$$\begin{aligned} \Sigma^{11} &= (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \\ &= [\Gamma_1' \Omega^{11} \Gamma_1 - (\Gamma_1' \Omega^{12} \Gamma_2) (\Gamma_2' \Omega^{22} \Gamma_2)^{-1} (\Gamma_2' \Omega^{21} \Gamma_1)]^{-1} \\ &= [\Gamma_1' (\Omega^{11} - \Omega^{12} (\Omega^{22})^{-1} \Omega^{21}) \Gamma_1]^{-1} \\ &= (\Gamma_1' \Omega_{11}^{-1} \Gamma_1)^{-1}. \quad \text{Q.E.D.} \end{aligned}$$

Proof of Theorem 1. Let $\{G_\gamma: \gamma \in V\}$ be the family of probability measures constructed in Lemma 1. If $\{G_\gamma: \gamma \in V\}$ satisfies the Local Asymptotic Normality (LAN) condition, then theorem 4.2 in Hájek (1972) provides the local asymptotic minimax bound, with $\{\sigma_{0,j,k}\}_{j,k=1}^p = J_0^{11}$, and Theorem 1 follows from Lemma 2. Sufficient conditions for LAN are given in conditions (A.1'), (A.2')(i), ..., (A.2')(iv) of Roussas (1972, p. 46; theorem 4.1, p. 53). We shall state these conditions in our notation and verify that they hold.

(A.1'): Let G_γ be the product measure: $G_\gamma^n(B_1 \times \cdots \times B_n) = \prod_{i=1}^n G_\gamma(B_i)$. We require that the probability measures $\{G_\gamma^n: \gamma \in V\}$ be mutually absolutely continuous for all $n \geq 1$. This holds since $\{\phi_1, \dots, \phi_r\}$ is the support of G_γ for all $\gamma \in V$.

(A.2')(i): Let $\pi_j(\gamma) = G_\gamma\{\phi_j\}$. Define $g(z; \gamma, \gamma^*)$: $Z \times V \times V \rightarrow R$ by $g(z; \gamma, \gamma^*) = \pi_j(\gamma^*)/\pi_j(\gamma)$ if $z = \phi_j$ for some $j \in \{1, \dots, r\}$; otherwise set $g(z; \gamma, \gamma^*) = g(\phi_1; \gamma, \gamma^*)$. We require that for each $\gamma \in V$, $f(\gamma, \gamma^*) = g^{1/2}(z; \gamma, \gamma^*)$ is differentiable in quadratic mean with respect to γ^* at the point (γ, γ) when z is distributed according to G_γ ; that is, if $D(z, \gamma)$ is the

derivative, then

$$\|h\|^{-2} \sum_{j=1}^r \pi_j(\gamma) \left[g^{1/2}(\phi_j; \gamma, \gamma + h) - g^{1/2}(\phi_j; \gamma, \gamma) - h'D(\phi_j, \gamma) \right]^2 \rightarrow 0$$

as $\|h\| \rightarrow 0$, where $\| \cdot \|$ is the usual Euclidean norm. This holds with

$$D(\phi_j, \gamma) = \frac{1}{2} \pi_j^{-1}(\gamma) \partial \pi_j(\gamma) / \partial \gamma,$$

since $\pi_j(\gamma)$ is continuously differentiable. We can set $D(z, \gamma) = D(\phi_1, \gamma)$ if $z \notin \{\phi_1, \dots, \phi_r\}$.

(A.2')(ii): We need to show that $D: Z \times V \rightarrow R^s$ is $\Psi \times \Delta$ -measurable, where Ψ and Δ are the σ -fields of Borel subsets of Z and V , respectively. Let B be a Borel subset of R^s ; then

$$\begin{aligned} D^{-1}(B) &= \{(z, \gamma) \in Z \times V: D(z, \gamma) \in B\} \\ &= \left(\bigcup_{j=1}^r \{\phi_j\} \times A_j \right) \cup (\tilde{Z} \times A_1), \end{aligned}$$

where $A_j = \{\gamma \in V: D(\phi_j, \gamma) \in B\}$ and $\tilde{Z} = Z - \{\phi_1, \dots, \phi_r\}$. Since $D(\phi_j, \cdot)$ is continuous, we have $A_j \in \Delta$ and so $D^{-1}(B) \in \Psi \times \Delta$.

(A.2')(iii): Note that

$$4E_{G_\gamma} [D(z, \gamma) D'(z, \gamma)] = J_\gamma.$$

We require that J_γ be positive-definite for all $\gamma \in V$. This follows from Lemma 1.

(A.2')(iv): For each fixed $\gamma \in V$, we require that $g(z; \gamma, \gamma^*)$ is $(\Psi \times \Delta)$ -measurable. Let B be a Borel subset of R . Then

$$\begin{aligned} g_\gamma^{-1}(B) &= \{(z, \gamma^*) \in Z \times V: g(z; \gamma, \gamma^*) \in B\} \\ &= \left(\bigcup_{j=1}^r \{\phi_j\} \times A_{\gamma j} \right) \cup (\tilde{Z} \times A_{\gamma 1}), \end{aligned}$$

where $A_{\gamma j} = \{\gamma^* \in V: g(\phi_j; \gamma, \gamma^*) \in B\}$ and $\tilde{Z} = Z - \{\phi_1, \dots, \phi_r\}$. Since $\pi_j(\cdot)$ is continuous, we have $A_{\gamma j} \in \Delta$ and so $g_\gamma^{-1}(B) \in \Psi \times \Delta$. Q.E.D.

Proof of Lemma 4. We can choose $\varepsilon > 0$ and F_0 -integrable functions $f_j: Z \rightarrow R$ such that $F \in \mathcal{T}$ if

$$\left| \int f_j(z) F(dz) - \int f_j(z) F_0(dz) \right| < \varepsilon, \quad j = 1, \dots, J.$$

Define $h_j(x) = \int f_j(x, y) F_0(dy|x)$. Now apply Lemma 3 to obtain a probability measure G_0^* , with support $\{\tau_1, \dots, \tau_l\} \subset X_0$, such that

$$\int D'_0(x) \Sigma_0^{-1}(x) D_0(x) G_0^*(dx) = \Lambda_0^{*-1},$$

and

$$\int h_j(x) G_0^*(dx) = \int h_j(x) F_0^*(dx), \quad j = 1, \dots, J.$$

For each x in the support of G_0^* , apply Lemma 3 again to obtain a probability measure $G_0(dy|x)$, whose support is a finite subset of Y , with

$$\int \rho(z, \theta_0) G_0(dy|x) = \int \rho(z, \theta_0) F_0(dy|x) = 0,$$

$$\int \rho(z, \theta_0) \rho'(z, \theta_0) G_0(dy|x) = \Sigma_0(x),$$

$$\int \frac{\partial \rho(z, \theta_0)}{\partial \theta'} G_0(dy|x) = D_0(x),$$

and

$$\int f_j(z) G_0(dy|x) = \int f_j(z) F_0(dy|x), \quad j = 1, \dots, J.$$

Now define $G_0(dx, dy) = G_0(dy|x) G_0^*(dx)$. Then

$$\int f_j(z) G_0(dz) = \sum_{k=1}^l G_0^*\{\tau_k\} \int f_j(\tau_k, y) G_0(dy|\tau_k) = \int f_j(z) F_0(dz),$$

which implies that $G_0 \in \mathcal{T}$. Q.E.D.

Proof of Proposition 2. Define

$$\begin{aligned} T_k &= E_{F_0} [\psi_k(z, \theta_0) \psi'_k(z, \theta_0)] \\ &= E_{F_0} [B_k(x) \Sigma_0(x) B'_k(x)]. \end{aligned}$$

T_k is positive-definite, for otherwise there is a non-zero $\phi \in R^q$ ($q = rk$) such that $B'_k(x)\phi = 0$ with F_0^* -probability one; but then $\{f_1, \dots, f_k\}$ is not linearly independent.

Let α be an arbitrary point in R^p and define $w(x) = D_0(x)\alpha$. Define

$$\Delta_k = \Lambda_0^{*-1} - \left[E_{F_0} \frac{\partial \psi_k(z, \theta_0)}{\partial \theta'} \right]' T_k^{-1} \left[E_{F_0} \frac{\partial \psi_k(z, \theta_0)}{\partial \theta'} \right],$$

and

$$\begin{aligned} \delta_k &= \alpha' \Delta_k \alpha \\ &= E_{F_0}(w' \Sigma_0^{-1} w) - E_{F_0}(w' B'_k) T_k^{-1} E_{F_0}(B_k w). \end{aligned}$$

We shall show that $\delta_k \rightarrow 0$ as $k \rightarrow \infty$. Since α is arbitrary and Δ_k is symmetric, this implies that $\Delta_k \rightarrow (0)$; hence $[E_{F_0} \partial \psi_k / \partial \theta']' T_k^{-1} [E_{F_0} \partial \psi_k / \partial \theta']$ is non-singular for k sufficiently large and $\Lambda_{0k} \rightarrow \Lambda_0^*$.

Consider the space of $r \times 1$ random vectors with inner product

$$\langle v_1, v_2 \rangle = \int v_1'(x) \Sigma_0^{-1}(x) v_2(x) F_0^*(dx),$$

and norm

$$\|v\| = \langle v, v \rangle^{1/2}.$$

Define

$$\beta_k^* = T_k^{-1} E_{F_0}(B_k w).$$

A simple calculation shows that

$$\delta_k = \|w - \Sigma_0 B'_k \beta_k^*\|^2,$$

and for any $\beta \in R^q$,

$$\|w - \Sigma_0 B'_k \beta\|^2 = \delta_k + \|\Sigma_0 B'_k (\beta_k^* - \beta)\|^2.$$

Hence

$$\delta_k = \min_{\beta \in R^q} \|w - \Sigma_0 B'_k \beta\|^2.$$

Define

$$u(x) = \Sigma_0^{-1}(x) w(x).$$

Then

$$\begin{aligned}\|w - \Sigma_0 B'_k \beta\|^2 &= \int (u - B'_k \beta)' \Sigma_0 (u - B'_k \beta) dF_0^* \\ &\leq \zeta \int (u - B'_k \beta)' (u - B'_k \beta) dF_0^* \\ &= \zeta \sum_{l=1}^r \int (u_l - \bar{f}'_k \beta_l)^2 dF_0^*,\end{aligned}$$

where u_l is the l th component of u and we have partitioned β into $\beta' = (\beta'_1, \dots, \beta'_r)$, where β_l is $k \times 1$. Since $u_l \in L_2(F_0^*)$ and $\{f_j\}$ is complete, given any $\varepsilon > 0$ we can choose β_l (for sufficiently large k) such that

$$\int (u_l - \bar{f}'_k \beta_l)^2 dF_0^* \leq \varepsilon / (r\zeta).$$

Then we have

$$\delta_k \leq \|w - \Sigma_0 B'_k \beta\|^2 \leq \varepsilon.$$

Since ε is arbitrary, we have shown that $\delta_k \rightarrow 0$. Q.E.D.

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